

Does FDA Enforcement Follow Product-Safety Risk?

FDA Regulatory Action Analysis · Team A06 · BA780

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Does enforcement go where the risk is?

94% of actions are tobacco warning letters to shops, so we set tobacco aside to see everything else.

37,623

Recalls

recall events, 2012–2026. Unsafe products pulled from sale.

177,252

Compliance Actions

actions, 2008–2026. The FDA disciplining a firm: warning letters, seizures and court injunctions.

1,724

The Link

facilities appear in both, joined by facility ID (FEI Number). This lets us test when the FDA responds.

Source: FDA Data Dashboard (Recalls; Compliance Actions)

How we built the exploratory analysis

1. Data Cleaning

Turned "-" into pandas missing values, dropped 30 rows with no facility ID, converted dates and added Year

2. Count recall events

Counted each Event ID once: 104,083 product rows became 37,623 recall events

3. Link by facility

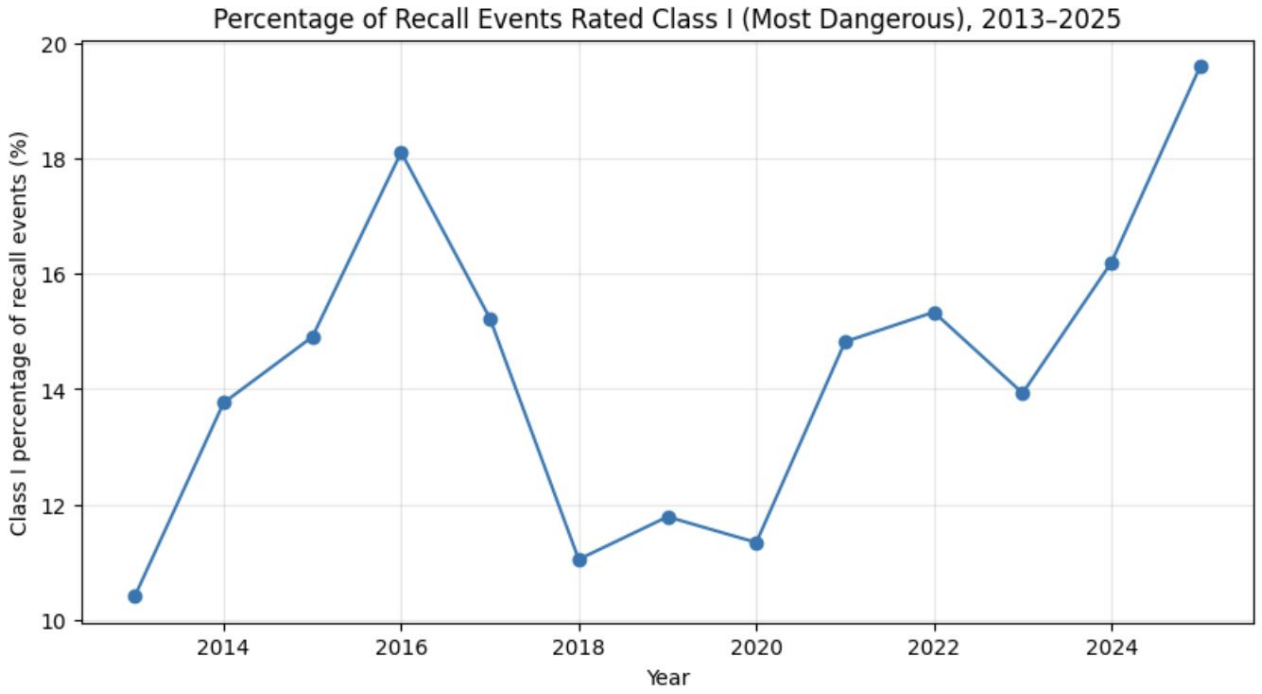
Joined recalls and actions by FEI Number into one table of 10,231 facilities

4. 15 questions

5 from the team assignment plus 10 deeper questions in 4 themes

4 themes: recall severity and causes · enforcement over time · facility-level risk · geography. Tobacco is excluded from comparisons because it is 94% of all actions.

Recalls are falling, but dangerous ones are rising



10.4%→ 19.6%

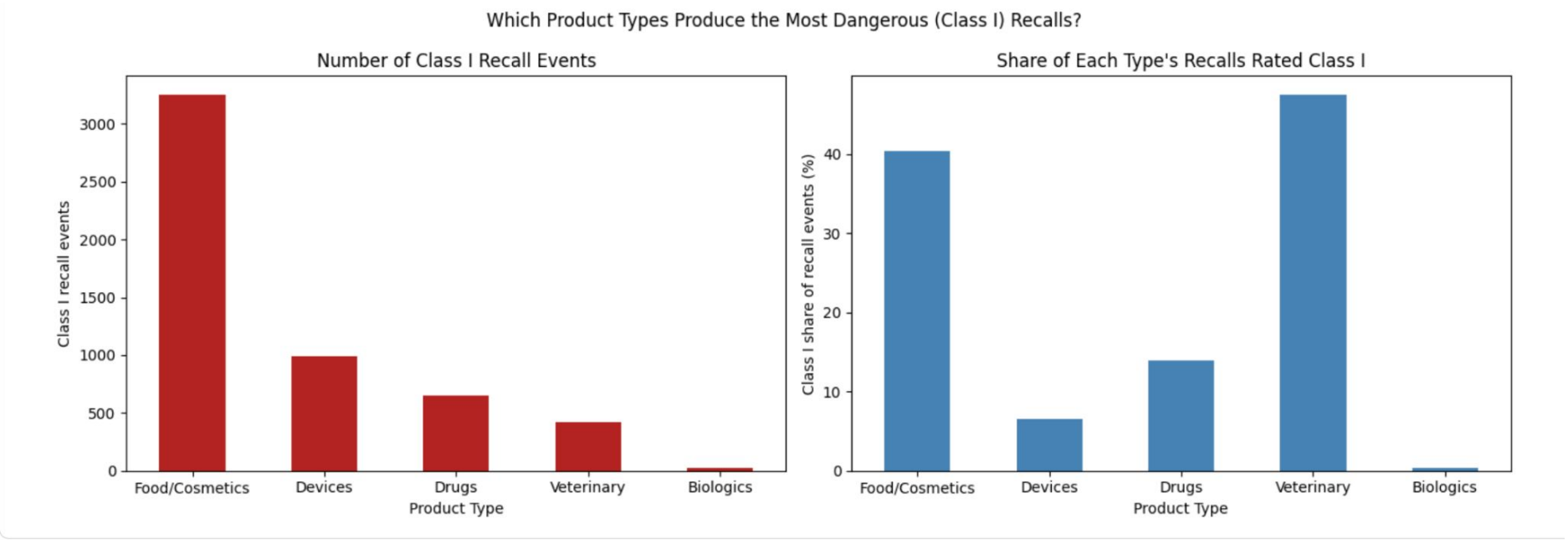
share of recall events rated Class I, 2013 to 2025

380 → 412

Class I recall events, while all recall events fell from 3,652 to 2,102

Notebook Q6

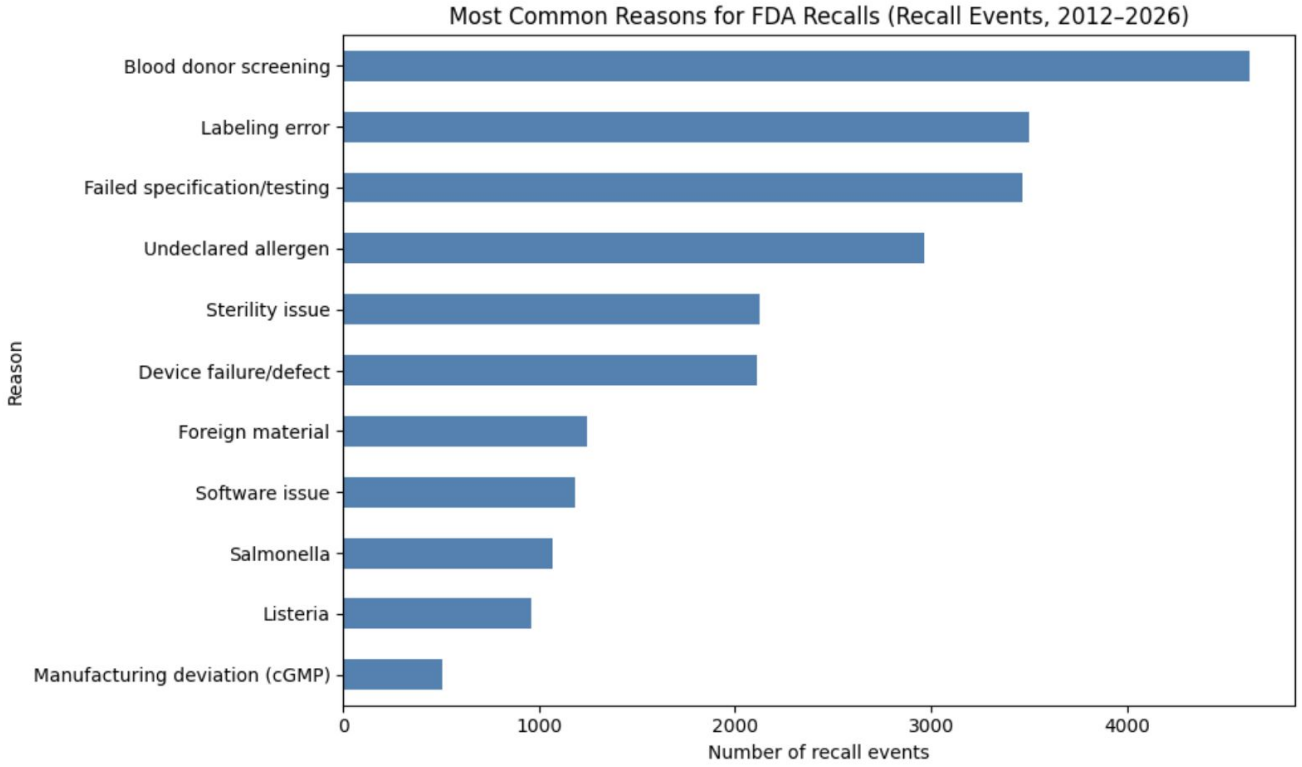
Food drives the most dangerous recalls



- **Food is high on both charts:** 3,251 Class I recalls (61% of all), and 40% of food recalls are dangerous (Class I).
- **Veterinary has the highest rate (47.5%) but few recalls (421);** devices are the opposite (6.6% rate, but 989 recalls).

Notebook Q7

Common causes aren’t the dangerous ones



3 top reasons

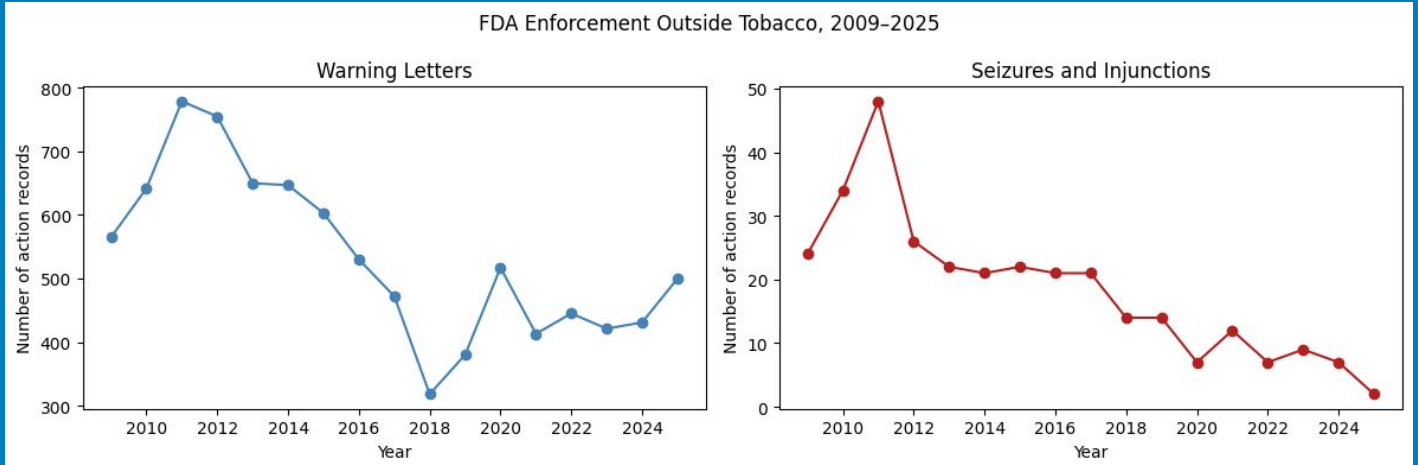
1. Blood donor screening
2. Labeling error
3. Failed specification

58%

Of Class I recalls were from Undeclared allergens, salmonella, and Listeria

Notebook Q8 9

Outside tobacco, enforcement has shrunk



48 → 2

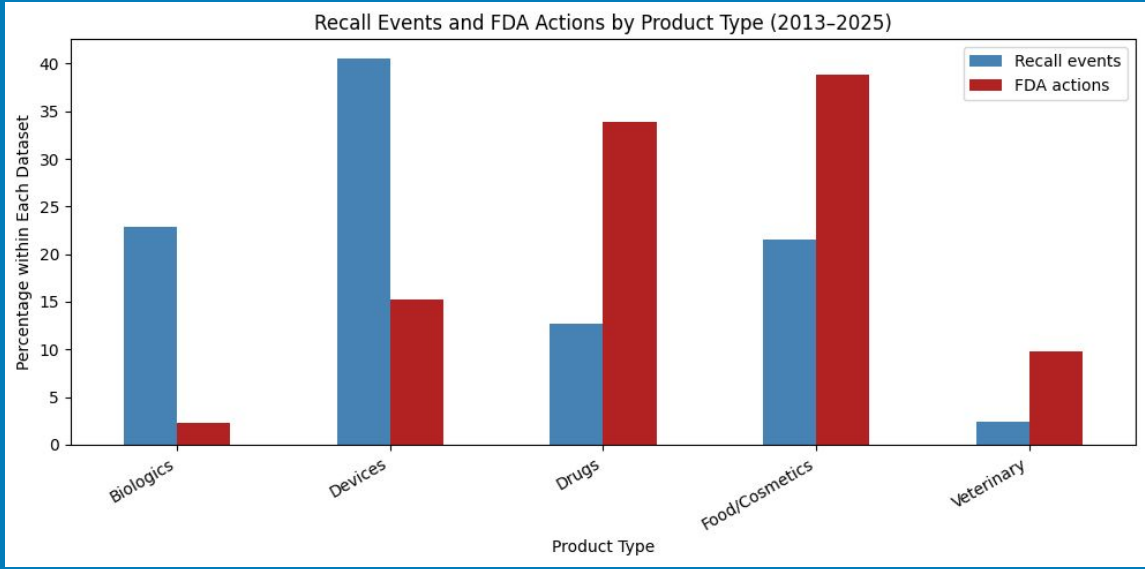
seizures and injunctions a year, 2011 to 2025

779 → 500

warning letters a year, 2011 to 2025

Notebook Q10

Actions don’t match where recalls are



57.6%

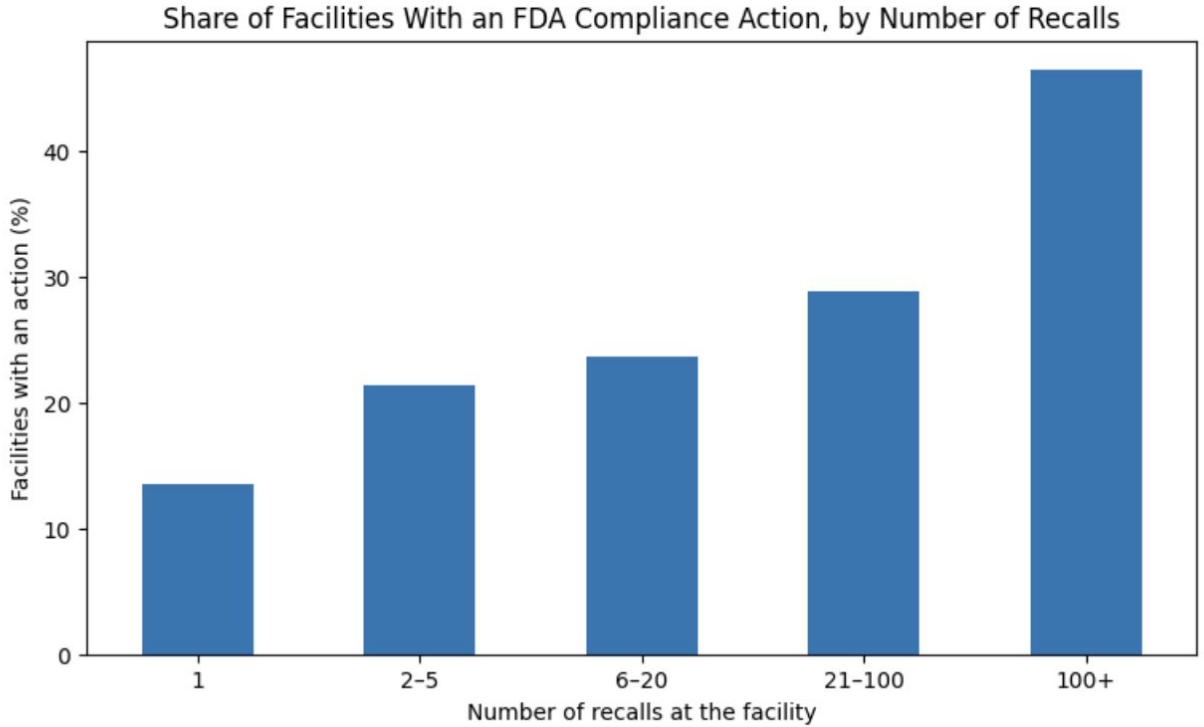
of seizures and injunctions go to food, the category with the most Class I recalls

Devices

2nd in Class I recalls, but only 3rd in FDA actions, behind drugs

Notebook Q2, Q5, Q7

FDA Action Rates Rise with Recall Frequency, While Severity Also Matters



13.5%→ 46.4%

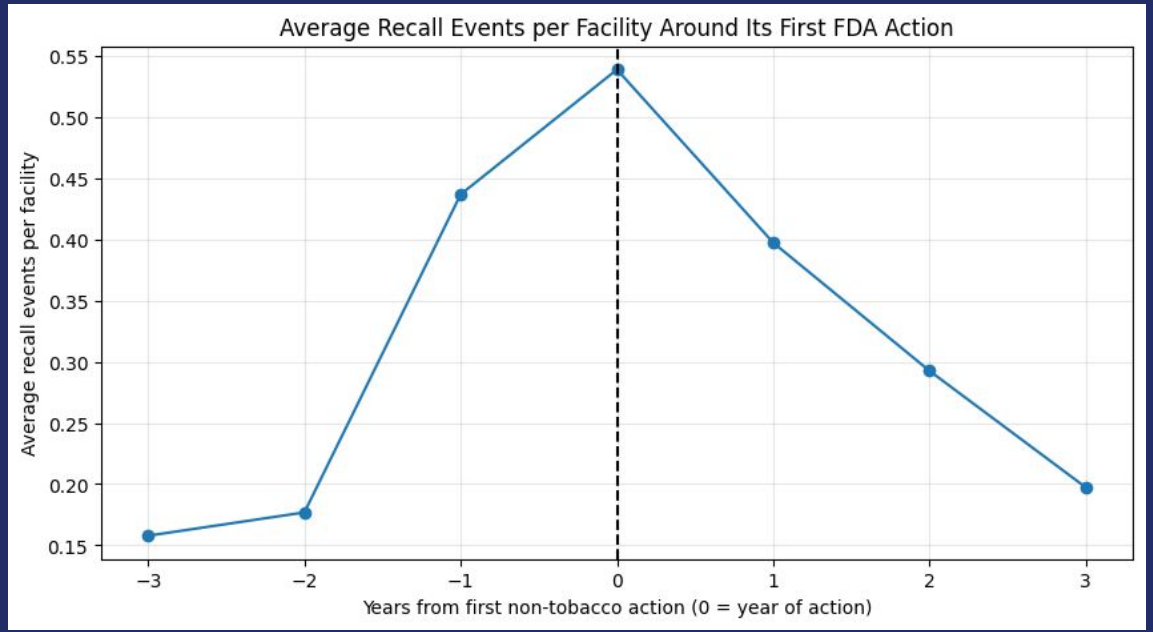
FDA action rates increase from facilities with one recall to those with 100+ recalls.

20% v.s. 15.3%

Action rates for facilities with a Class I recall versus those with only lower-severity recalls.

Notebook Q11 12

Actions come after the spike, not before



0.16 → 0.54

recall events per facility, from 3 years before the first action to the action year

-63%

drop from that peak over the next 3 years

The FDA reacts once problems peak. Acting at the first rise could prevent the spike.

Notebook Q13

Labeling & allergens lead recall problems

Labeling / allergens

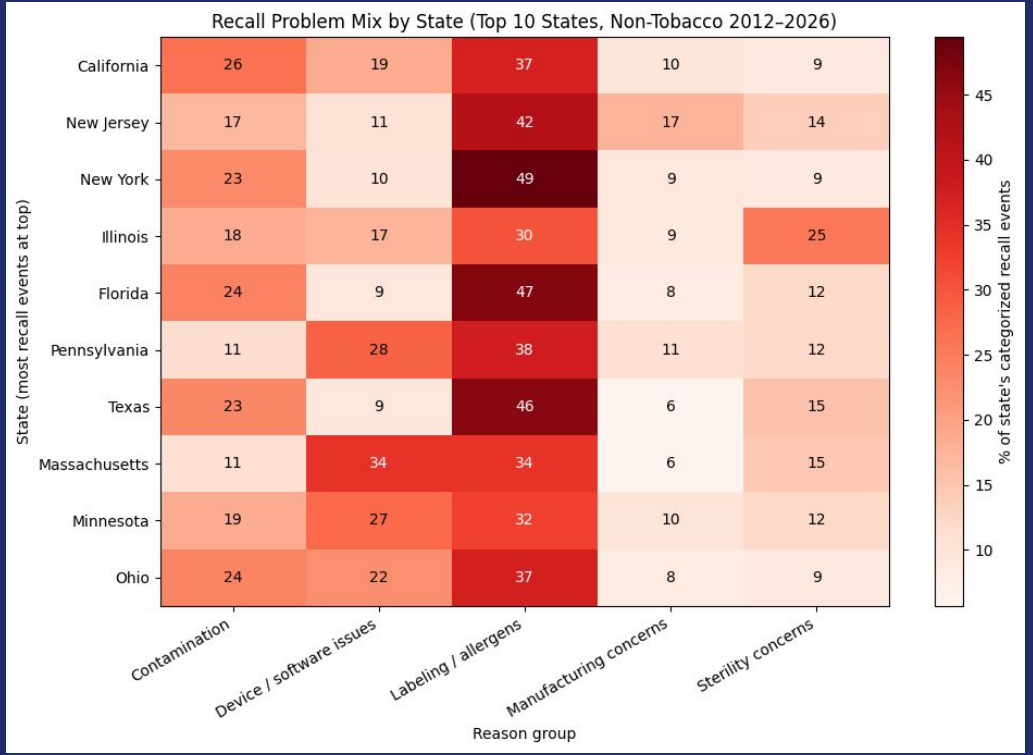
New York: **49%**

Florida: **47%**

Device / software issues

Massachusetts: **34%**

Pennsylvania: **28%**



Notebook Q14

Some states lag on enforcement

Virginia

Repeat Rate: **48.6%**

Action Rate: **14.7%**

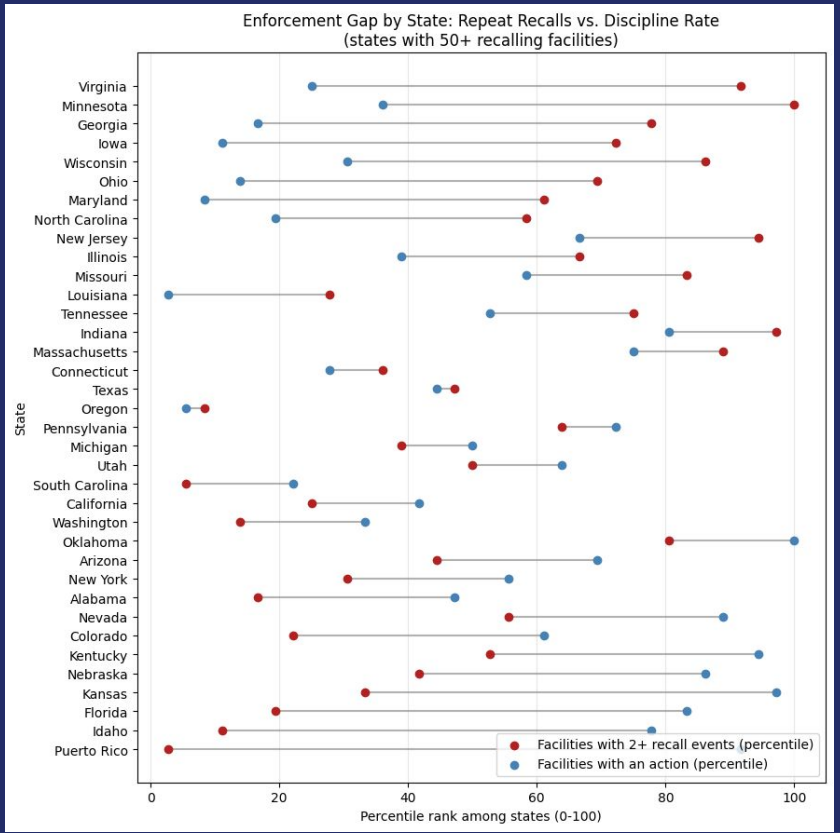
66.7
Percentile Point Gap

Minnesota

Repeat Rate: **50.0%**

Action Rate: **15.6%**

63.9
Percentile Point Gap



Notebook Q15

Our Recommendation

1. Early Warning Score

Flag facilities with rising or Class I recalls, so action comes before the peak, not after it.

2. Food Pathogens & Allergens

They cause 58% of Class I recalls, so focus food inspections on allergen labeling and pathogen testing.

3. Device Oversight

Devices are 40.5% of recall events but only 15.2% of actions; review repeat device recallers.

4. Repeat Recallers

53.6% of facilities with 100+ recall events were never disciplined; start there, and in Virginia and Minnesota.

Q&A